International Journal of Mathematical Modeling, Simulation and Applications
ISSN 0973-8355 Vol. 3 No. 4, 2010 pp. 446-454

Radiation and Darcy Effects on Unsteady MHD Heat and Mass Transfer Flow of a Chemically Reacting Fluid Past an Impulsively Started Vertical Plate

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ABSTRACT This paper is focused on the effect of thermal radiation on the natural convective heat and mass transfer of a viscous, incompressible, electrically conducting and chemically reacting fluid past an impulsively started moving vertical plate adjacent to Darcian porous regime. The dimensionless governing equations are solved using an implicit finite-difference method of Crank-Nicholson type. A parametric study is performed to illustrate the influence of thermophysical parameters on the velocity, temperature and concentration profiles. Also, the local and average skin-friction, Nusselt number and Sherwood number are presented graphically. It is found that thermal radiation reduces both velocity and temperature in the boundary layer. Increasing the Darcy number is seen to accelerate the flow. An increase in the chemical reaction parameter leads a fall in the concentration. This model finds applications in solar energy collection systems, porous combustion and transport in fires in porous media (forest fires) and also in the design of high temperature chemical processing systems.

Keywords: Radiation, MHD, Darcy, chemically reacting fluid

1. INTRODUCTION

In the processes involving high temperatures, the radiation heat transfer in combination with conduction, convection and also mass transfer plays very important role in the design of pertinent equipment in the areas such as nuclear power plants, gas turbines and the various propulsion devices for aircraft, missiles, satellites and space vehicles. Chamkha et al. [1] studied the radiation effects on the free convection flow past a semi-infinite vertical plate with mass transfer. Muthucumaraswamy and Kumar Senthil [2] investigated the heat and mass transfer effect on moving vertical plate in the presence of thermal radiation. Ramachandra Prasad et al. [3] considered the radiation and mass transfer effects on two-dimensional flow past an impulsively started isothermal vertical plate. The interaction of radiation with hydromagnetic flow has become industrially more prominent in the processes wherever high temperatures occur. Chaudhary et al. [4] studied the radiation effects with simultaneous thermal and mass diffusion in MHD mixed convection flow from a vertical surface. Emad M. Aboeldahad and Gamal El-Din A. Azzam